Lab 3: Modelling the Cell Membrane Potential

Stream 1: Cell Signalling

August 21, 2026

1 Goal

To model the cell membrane potential using ordinary differential equations (ODEs) based on ion channel dynamics, solve the system numerically, and analyse how the potential reaches a steady state (the resting potential) and responds to active transport mechanisms like the Na^+/K^+ pump.

2 Reading material

Keener, J., & Sneyd, J. (2025). *Mathematical Physiology* (3rd ed.). Springer.

- **Chapter 2: Cellular Homeostasis**, particularly Section 2.5 on Membrane Potential.

3 Guided example: a simple model of resting potential

You will build a model for the membrane potential (V) based on the passive flow of potassium (K^+) and sodium (Na^+) ions through leak channels. The script `*.ipynb` in ED Resources should be structured to match the steps below.

The membrane potential dynamics are described by the equation:

$$C_m \frac{dV}{dt} = -I_{\text{ion}} = -(I_K + I_{Na}) \tag{1}$$

where C_m is the membrane capacitance and the ionic currents are modeled as:

$$I_K = g_K(V - E_K)$$
$$I_{Na} = g_{Na}(V - E_{Na})$$

Here, g_K and g_{Na} are the constant conductances for the leak channels, and E_K and E_{Na} are the Nernst potentials for each ion. Use the following physiological parameters:

- $C_m = 1 \quad (\mu\text{F/cm}^2)$
- $g_K = 0.3 \quad (\text{mS/cm}^2)$
- $g_{Na} = 0.015 \quad (\text{mS/cm}^2)$
- $E_K = -75 \quad (\text{mV})$
- $E_{Na} = 55 \quad (\text{mV})$

Part A: solving the ODE and plotting the potential

First, simulate how the membrane potential evolves over time from an initial state.

1. **Define the System in Python:** Create a function for the ODE (1) that takes time, the state variable (V), and the parameters.
2. **Solve Numerically:** Use the `compute` function to integrate the system from an initial condition of $V(0) = 0$ mV and simulate for 50 ms
3. **Plot the Results:** Generate a time series plot showing how $V(t)$ evolves and approaches its steady state.

Part B: analysing the steady state

Next, analyse the system's equilibrium.

1. **Calculate the resting potential:** The steady state, or resting potential (V_{rest}), occurs when $\frac{dV}{dt} = 0$. Solve this condition ($I_K + I_{Na} = 0$) analytically for V.
2. **Compare results:** Compare the steady-state value from your numerical simulation in Part A to your analytical calculation. They should match.
3. **Plot the I-V curve:** Create a plot of the total ionic current, I_{ion}, as a function of the membrane potential, V. The voltage at which the curve crosses the x-axis ($I_{\text{ion}} = 0$) is the resting potential. This plot is fundamental in electrophysiology.

4 Exercise: the role of the Na^+/K^+ pump

Now, let's make the model more realistic by including the Na^+/K^+ pump, which actively transports ions to counteract the passive leaks and maintain concentration gradients. The pump contributes a constant outward current, I_p. The system equation becomes:

$$C_m \frac{dV}{dt} = -(I_K + I_{Na} + I_p) \tag{2}$$

Assume the pump current is $I_p = 0.5\,(\mu\text{A/cm}^2)$.
(Note: By convention, a positive sign indicates a net outward flow of positive charge).

Task: analyse the membrane potential with the pump

Your task is to adapt the Python script from the guided example to analyse the model including the pump.

1. **Modify the code:**
   - Update the ODE function to implement equation (2).
   - Add the new parameter I_p to your script.
2. **Generate plots and analyse:**
   - Simulate the system again, starting from $V(0) = 0$ mV, and plot the new time course of $V(t)$.
   - Analytically solve for the new resting potential when $\frac{dV}{dt} = 0$ and compare it to your simulation.
   - Plot the new total current ($I_K + I_{Na} + I_p$) versus voltage. How has the I-V curve shifted compared to the one from the guided example?
3. **Discussion:** Based on your results, what is the qualitative effect of the Na^+/K^+ pump on the resting membrane potential? Why is this physiologically important? Discuss with your peers!

Advanced task: threshold behavior and bifurcation

Let's analyse the behaviour of the excitable membrane model. The system equation is:

$$C_m \frac{dV}{dt} = -\left(g_K(V - E_K) + g_{Na}(V)(V - E_{Na})\right) + I_{app} \tag{3}$$

where the voltage-dependent sodium conductance is given by:

$$g_{Na}(V) = \bar{g}_{Na}\left(1 + \tanh\left(\frac{V - V_{th}}{V_{slope}}\right)\right)$$

For this task, use the following parameters: $\bar{g}_{Na} = 0.5$, $V_{th} = -20$ mV, $V_{slope} = 5$ mV, and a potassium leak conductance of $\mathbf{g_K = 1.1\ mS/cm^2}$.

Part 1 (PyDSTool): continuation analysis

1. **Set up the PyCont instance:** Create a 'EP-C' curve and with the initial value of the applied current, `Iapp`, 0.
2. **Run continuation:** Find a steady state using the `find_fixedpoints` function and run the continuation in the `forward` direction.
3. **Analyse the bifurcation diagram:** The diagram shows the steady-state voltage as a function of the applied current. You should observe a region where the system has three steady states (bistability).

Discussion

- From your diagrams, identify the approximate values of the applied current for both saddle-node bifurcation points.
- What is the physiological significance of the lower stable branch versus the upper stable branch? What do these two states represent in a neuron?
- Imagine that you slowly increase the applied current from 0. Describe what happens to the steady-state membrane potential V. At what approximate current value does a dramatic change occur? Discuss with your peers!